Lab Session N°1

Introduction to JADE, Installation & First Agent
Problem Solving & Configuration

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1 Theoretical Background: Agents & JADE

1.1 What is a Software Agent?

An agent is a computer system that is situated in some environment and that is capable of autonomous action in this environment in order to meet its design objectives. Key properties include:

- **Autonomy:** Operates without direct human intervention.
- **Social Ability:** Interacts with other agents via an Agent Communication Language (ACL).
- **Reactivity:** Perceives and responds to changes in the environment.
- **Proactivity:** Exhibits goal-directed behavior.

1.2 The JADE Framework

JADE (Java Agent DEvelopment Framework) is a software framework fully implemented in Java. It simplifies the implementation of multi-agent systems by providing a FIPA-compliant middleware.

Key JADE Architecture Concepts:

- **Agent Platform:** The environment where agents live.
- **Container:** A runtime environment (a Java Virtual Machine instance) that hosts agents.
- **Main Container:** The first container to start. It holds two special agents:
  - **AMS (Agent Management System):** The authority that manages the platform (naming, creation, deletion).
  - **DF (Directory Facilitator):** The "Yellow Pages" where agents advertise their services.

2 Step 1: Environment Setup

1. **Download JADE:**
   - Go to the official website: `https://jade.tilab.com/`.
   - Download the latest version (e.g., `JADE-bin-4.6.0.zip`).
2. **Extract Files:**
   - Unzip the file. Locate the `lib` folder. It contains `jade.jar` and `commons-codec-x.x.jar`.

3 Step 2: Eclipse Project Configuration

1. Open Eclipse and create a new Java Project named **TP1_JADE**.
2. Right-click on the project → **Build Path** → **Configure Build Path**.
3. Go to the **Libraries** tab (select *Classpath* if available).
4. Click **Add External JARs** and select `jade.jar` and `commons-codec.jar` from your JADE `lib` folder.
5. Click *Apply and Close*.

CRITICAL CONFIGURATION (Fixing Java Versions)

Modern Java versions (Java 9+) use a Module system that is incompatible with JADE. You must perform this step to avoid the error:
"Error: Could not find or load main class jade.Boot in module..."

1. Check your project's `src` folder.
2. Look for a file named `module-info.java`.
3. **DELETE this file.**

Deleting this file forces Eclipse to treat the project as a classic Java application, which is required for JADE.

4 Step 3: Creating the "Hello World" Agent

1. Create a package: `eu.uemf.agents`. 2. Create a class: `HelloAgent`. 3. Paste the following code:

```
package eu.uemf.agents;

import jade.core.Agent;

public class HelloAgent extends Agent {

    // The setup() method is the entry point (like main)
```

```java
    protected void setup() {
        System.out.println("Hello World! I am a JADE Agent.");

        // Print the Agent's Local Name
        System.out.println("My local name is " + getAID().
           getLocalName());

        // Print the Agent's Globally Unique ID (GUID)
        System.out.println("My GUID is " + getAID().getName());
    }
}
```

Listing 1: HelloAgent.java

5 Step 4: Running the Platform (Resolving Network Errors)

To run a JADE agent, we do not run the agent class directly. We run the **JADE Boot** loader.

1. Go to **Run** → **Run Configurations...**
2. Create a new **Java Application**.
3. **Main Class:** `jade.Boot`
4. **Program Arguments:** Copy the line below EXACTLY.

The Arguments (Fixing "No ICP Active")

Many systems fail with "Communication failure" or "No ICP active" because JADE cannot detect the correct IP address. We must force it to use `localhost`.
Use these arguments:

```
-gui -local-host 127.0.0.1 -agents myAgent:eu.uemf.agents.HelloAgent
```

Explanation:

- `-gui`: Launches the Graphical User Interface (RMA).
- `-local-host 127.0.0.1`: Forces JADE to bind to the local loopback address (Fixes connection errors).
- `-agents name:Class`: Tells JADE to create this agent immediately.

6 Step 5: Execution & Screenshots

Click **Run**. If configured correctly, you should see two results.

6.1 Console Output

You should see the "Hello World" message and the GUID of the agent.

```
INFO: Service jade.core.event.Notification initialized
Feb 03, 2026 10:57:35 AM jade.mtp.http.HTTPServer <init>
INFO: HTTP-MTP Using XML parser com.sun.org.apache.xerces.internal.jaxp.SAXParserImpl$JAXPSAXParser
Feb 03, 2026 10:57:35 AM jade.core.messaging.MessagingService boot
INFO: MTP addresses:
http://waga-Vostro-15-3530:7778/acc
Bonjour ! Je suis un agent JADE.
Feb 03, 2026 10:57:35 AM jade.core.AgentContainerImpl joinPlatform
INFO: ----------------------------------------
Agent container Main-Container@127.0.0.1 is ready.
--------------------------------------------
Mon nom local est : monPremierAgent
Mon GUID est : monPremierAgent@127.0.0.1:1099/JADE
```

Figure 1: Eclipse Console Output showing the agent's print statements.

6.2 JADE GUI (Remote Agent Management)

The graphical interface should appear, showing the "Main-Container" and your agent.

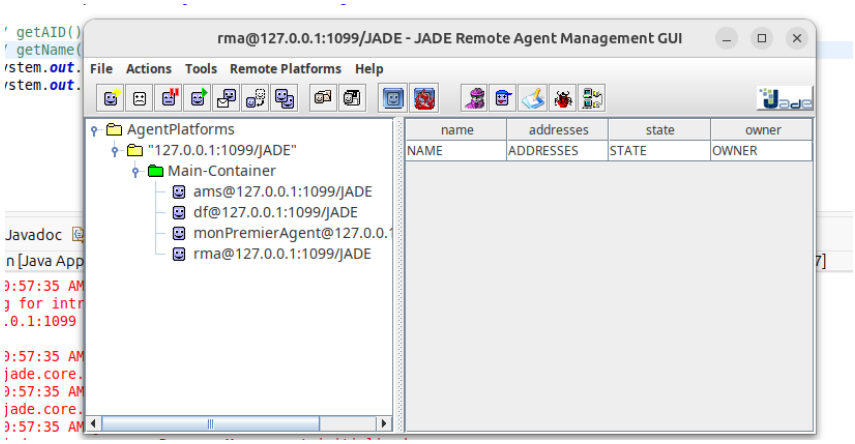

Figure 2: The JADE GUI showing the Main Container and 'myAgent'.